

A counterexample to conjecture 00000001676: the isoperimetric profile of the torus grid

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Abstract

Conjecture 00000001676 asserts that the isoperimetric profile of the $n \times n$ torus grid $C_n \square C_n$ is $\text{ip}(m) = \lceil 4\sqrt{nm - m^2} \rceil$ for all $m \leq n^2/2$. We refute the conjecture. It has two independent defects. First, on part of its stated range the radicand $nm - m^2$ is negative, so the formula is not even real: for $n = 3$, $m = 4$ one has $m \leq n^2/2 = 4.5$ but $nm - m^2 = -4$. Second, wherever it is defined the formula is wrong. The cleanest counterexample is $(n, m) = (3, 3)$: the formula gives $\lceil 4\sqrt{9 - 9} \rceil = 0$, whereas the true minimum edge boundary is 6.

1 Definitions

Definition 1 (Torus grid). Let C_n be the cycle graph on n vertices. The $n \times n$ *torus grid* is the Cartesian product $C_n \square C_n$. Its vertex set is $\mathbb{Z}_n \times \mathbb{Z}_n$, and two vertices are adjacent when they differ in exactly one coordinate, differences being taken modulo n (the torus wrap-around). The graph is 4-regular and has n^2 vertices and $2n^2$ edges.

Definition 2 (Edge boundary and isoperimetric profile). For $S \subseteq V(C_n \square C_n)$ the *edge boundary* is

$$\partial S = \{ \{u, v\} \in E : |\{u, v\} \cap S| = 1 \}, \quad |\partial S| = \sum_{v \in S} (4 - |N(v) \cap S|).$$

The *isoperimetric profile* is $\text{ip}(m) = \min \{ |\partial S| : S \subseteq V, |S| = m \}$.

Definition 3 (Conjectured formula). (*Conjecture 00000001676.*) For all $m \leq n^2/2$,

$$\text{ip}(m) = \lceil 4\sqrt{nm - m^2} \rceil,$$

attained by diagonal cuts, with no other minimizing configuration.

2 Exhaustive computation of the true profile

For $n = 3$ and $n = 4$ we computed $\text{ip}(m)$ by brute force: enumerate every m -vertex subset of the n^2 vertices, evaluate $|\partial S|$, and take the minimum. Subsets were encoded as bitmasks and ∂S was evaluated as $\sum_{v \in S} \#\{w \in N(v) : w \notin S\}$, which counts every crossing edge once. For $C_3 \square C_3$ there are $2^9 = 512$ masks; for $C_4 \square C_4$, $2^{16} = 65536$ masks. The results are in Table 1. (The script `reproduce.py` regenerates them; the Lean development in `lean4/` verifies the decisive finite cases by kernel evaluation over the 512 masks.)

$$C_3 \square C_3, \text{ stated range } m \leq 9/2, \text{ i.e. } m = 1, \dots, 4$$

m	$\text{ip}(m)$ true	formula	$nm - m^2$	status
1	4	$\lceil 4\sqrt{2} \rceil = 6$	2	formula wrong
2	6	6	2	agrees
3	6	$\lceil 4\sqrt{0} \rceil = 0$	0	formula wrong
4	8	undefined	-4	not real

$$C_4 \square C_4, \text{ stated range } m \leq 8$$

m	$\text{ip}(m)$ true	formula	$nm - m^2$	status
1	4	$\lceil 4\sqrt{3} \rceil = 7$	3	formula wrong
2	6	8	4	formula wrong
3	8	7	3	formula wrong
4	8	$\lceil 4\sqrt{0} \rceil = 0$	0	formula wrong
5	10	undefined	-5	not real
6	10	undefined	-12	not real
7	10	undefined	-21	not real
8	8	undefined	-32	not real

Table 1: True minimum edge boundary versus the conjectured formula.

3 The counterexample

Theorem 4 (Cleanest counterexample). *For the torus grid $C_3 \square C_3$ and $m = 3$, the conjectured formula of Definition 3 gives 0, but the true minimum edge boundary is 6. Hence the conjecture is false.*

Proof. Here $n = m = 3$, so $nm - m^2 = 9 - 9 = 0$ and the formula gives $\lceil 4\sqrt{0} \rceil = 0$.

A full row $\{(i, 0), (i, 1), (i, 2)\}$ of the torus has 3 vertices. Each of its vertices has its two horizontal neighbours inside the row and its two vertical neighbours outside the row, so the induced subgraph has 3 edges and

$$|\partial S| = 3 \cdot 4 - 2 \cdot 3 = 12 - 6 = 6.$$

The exhaustive search over all $\binom{9}{3} = 84$ triples (Table 1) shows no triple has a smaller boundary, so $\text{ip}(3) = 6$. Thus the conjecture predicts 0 where the truth is 6. $\square$

Proposition 5 (Further defined-range failures). *The formula is also wrong at $(n, m) = (3, 1)$: it gives $\lceil 4\sqrt{2} \rceil = 6$, while a single vertex has boundary 4. More generally Table 1 exhibits six distinct defined values of (n, m) in the stated range at which the formula disagrees with the true profile.*

4 Ill-posedness of the stated range

The conjecture quantifies over all $m \leq n^2/2$. But the radicand $nm - m^2 = m(n - m)$ is negative precisely when $m > n$. Together with the upper bound $m \leq n^2/2$ this is non-empty for every $n \geq 3$: choose any integer m with $n < m \leq n^2/2$ (for $n = 3$, $m = 4$; for $n = 4$, $m = 5, 6, 7, 8$; in general such an m exists because $n^2/2 \geq n + 1$ for $n \geq 3$). At those points the formula is not a real number, so the conjecture is not even well posed on its stated domain.

Remark 6. For $n = 3$, $m = 4$: $nm - m^2 = 12 - 16 = -4 < 0$, while $m = 4 \leq n^2/2 = 4.5$. For $n = 4$, $m = 8$: $nm - m^2 = 32 - 64 = -32 < 0$, while $m = 8 = n^2/2$. Thus the defect occurs within the stated range, not merely beyond a boundary convention.

5 Verdict

The conjecture is **false**, on two independent grounds:

1. **Ill-posed on part of its range.** For $n < m \leq n^2/2$ the radicand $nm - m^2$ is negative (e.g. $n = 3$, $m = 4$), so the formula is not real.
2. **Wrong wherever defined.** The cleanest instance is $(n, m) = (3, 3)$: the formula gives 0 while the true minimum edge boundary is 6. Other defined-range failures include $(n, m) = (3, 1)$ and $(4, 1), (4, 2), (4, 3), (4, 4)$.